electrons would fall into nuclei, there would be no atoms and we would not exist.

The same applies to the universe. Let us imagine a universe contracting and becoming extremely small, squashed by its own weight. According to Einstein's equations, this universe would be squashed ad infinitum and at a certain point would disappear altogether, like the electron falling into the nucleus. This is the Big Bang predicted by Einstein's equations, if we ignore quantum theory.

But if we take quantum mechanics into account, the universe cannot be indefinitely squashed. A quantum repulsion makes it rebound. A contracting universe does not collapse down to a point: it bounces back and begins to expand, as if it were emerging from a cosmic explosion (figure 8.3).

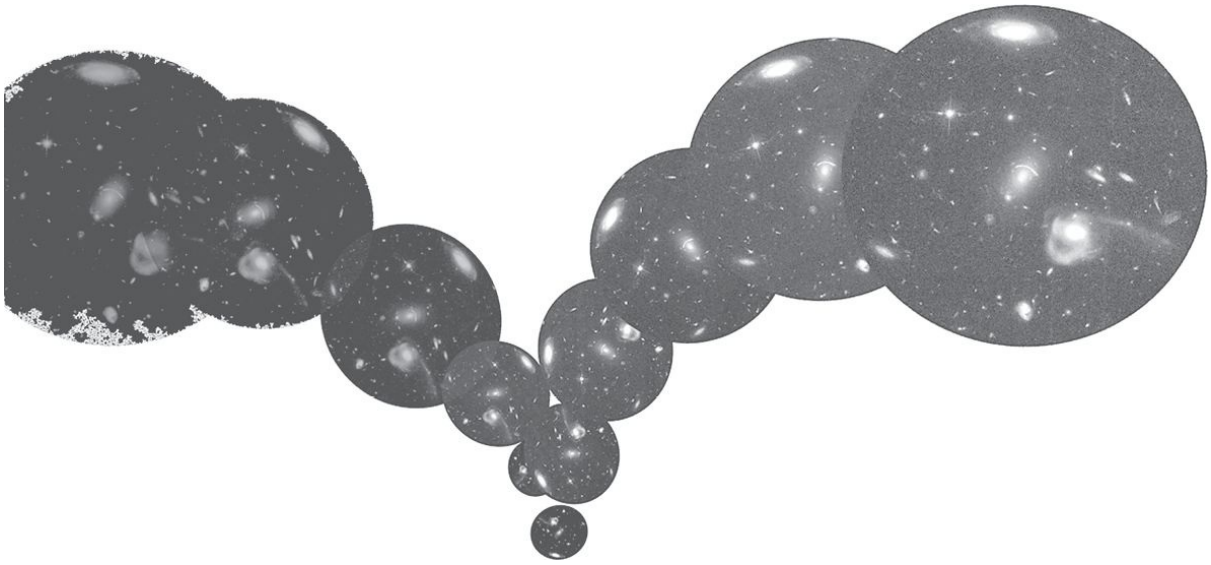


Figure 8.3 The Big Bounce of the universe in a graphical representation by Francesca Vidotto, the Italian scientist who first used spinfoams to compute the probability of this process.

The past of our universe may therefore well be the result of just such a rebound. A gigantic rebound known as a Big Bounce instead of Big Bang. This is what seems to emerge from the equations of loop quantum gravity when they are applied to the expansion of the universe.

The image of the bounce must not be taken literally. Going back to the example of the electron, recall that if we want to place an electron as close as possible to an atom, the electron is no longer a particle; we can think of it, instead, as opened up in a cloud of probabilities. An exact position no longer makes sense for the electron. The same for the universe: in the crucial passage through the Big Bounce, we can no longer think of a single, although granular, space and time, but only of a spread-out cloud of probabilities in which time and space wildly fluctuate. At the Big Bounce, the world is